



POLITECNICO MILANO 1863

A.Y. 2024-2025 Software Engineering (for Automation)

PROJECT GUIDELINE

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1 Goal and approach

The objective of this project is to apply in practice what you learn during lectures with the purpose of becoming familiar with software engineering practices. The project includes three deliverables:

1. **Requirement Analysis and Specification Document** for a problem provided by students and approved by the instructor.
2. The definition of the Design specified in a **Design Document** for the system considered in point 1 above.
3. The **software implementation** consistent with requirements and design specified by the documents in point 1 and 2, respectively.

Deliverables will be reviewed during the final discussions that will take place after submission of the three. The evaluation will assess the quality of the artifacts (precision, conceptual and technical soundness, completeness, consistency) and the quality of your presentation (if you are able to explain your point in an appropriate way and if your presentation fits in the allowed time). The three deliverables are described in the rest of this document.

2 Project schedule

Students interested in opting for this exam mode are required to submit a project proposal—a concise, 1-page document outlining the scope and problem they intend to address. While there is **no strict deadline** for submitting proposals, it is highly recommended that students do so by the conclusion of the lecturing period. This ensures ample time for evaluation and, if approved, allows for the recording

of scores by the end of the academic year.

IMPORTANT: upon the approval of the project proposal, students decide along with the instructor a starting date. From such a date, students have **3 months** to deliver the three specified artifacts.

3 Rules

- This assignment is optional and replaces the written exam.
- The project is developed in groups of 2 or 3 persons. Groups composed of a single student are not allowed. The project and the corresponding expectations of the professor will be calibrated based on the size of the group.
- Each group interested in taking the project must send a project proposal as written above (matteo.camilli@polimi.it).
- Each group must provide the requested artifacts within 3 months after the starting date. A delay, if notified in advance, will be tolerated, but it may result in a penalty in the final score. These artifacts will be presented in a final meeting that will be scheduled on due time.
- Each group must release artifacts by committing and then pushing into the main branch of a git repository created for the project (see the following section).
- The material included in your artifacts is not fixed in stone. You can (and are encouraged to) provide updates at any point before the final submission if you think these are needed.

4 Group organization

Students form groups and register by following these steps:

1. Create a private repository for your project on GitHub (<https://github.com>). Please note that, as students, you have the possibility to create a private GitHub repository for free. Your repository should be named by combining the names of all group members (e.g., SE4A_MatteoCamilli_LiviaLestingi). Make sure that all group members have a Github account and have access to the repository (write permission). Moreover, invite the professor (GitHub account *matteocamilli*) to access your repository.
2. Create a directory for each of the deliverables you will be working on.

5 Deliverables

The first 2 deliverables are documents you produce will include the following elements:

- **A FRONT PAGE** that includes the project title, the version of the document, your names and the release date.
- **A TABLE OF CONTENTS** that includes the headers of the first three levels of headings in your document, with the corresponding page number. At the beginning of this document, you find a table of contents that you can use as an example.

The specific characteristics of each document are described in the next subsections.

5.1 Deliverable 1 – Requirements Analysis and Specification Document

The *Requirements analysis and specification document* contains the description of the scenarios, the domain, the requirements, and the models describing the requirements and specification for the problem under consideration. As a reference structure for your document, you should refer to the one reported below.

1. **INTRODUCTION**
 - A. Purpose: goals of the project
 - B. Scope: high-level description of the world and the shared phenomena
 - C. Definitions, Acronyms, Abbreviations
 - D. Document Structure
2. **OVERAL DESCRIPTION**
 - A. Scenarios: informal description of relevant scenarios to illustrate the shared phenomena between the system and the external world (including all the relevant actors)
 - B. Domain model: class diagram describing the domain elements and corresponding relationships
 - C. User characteristics: anything that is relevant to clarify their needs
 - D. Product functions: most important requirements
 - E. Nonfunctional aspects: most important nonfunctional requirements
 - F. Assumptions, dependencies and constraints
3. **ADDITIONAL MODELS**
 - A. Requirements-level sequence diagrams: further details about requirements (if needed)
 - B. Finite State Machines: further details about requirements (if needed)
4. **REFERENCES**

5.2 Deliverable 2 – Design Document

The *Design document* must contain the description of the system architecture (both structural and behavioral views). You should use all the UML diagrams you need to provide a full description of the system. As a reference structure for your document please refer to the following one.

1. **INTRODUCTION**
 - A. Purpose: recap of the main goals of the project
 - B. Definitions, Acronyms, Abbreviations
 - C. Revision history
 - D. Document Structure
2. **ARCHITECTURAL DESIGN**
 - A. Component view: component diagram, brief description of components and main operations exported by interfaces
 - B. Class view: further details on individual components through class diagrams (if relevant)
 - C. Runtime view: sequence diagrams to describe the way components (or objects) interact to accomplish specific tasks typically related to scenarios/requirements
 - D. Selected architectural styles and patterns: Please explain which styles/patterns you used, why, and how

3. **USER INTERFACE DESIGN:** Provide an overview on how the user interface(s) of your system will look like.
4. **REQUIREMENTS TRACEABILITY:** show and explain how the requirements you have defined in the requirements analysis document map to the design elements defined in this document
5. **IMPLEMENTATION, INTEGRATION AND TEST PLAN:** Identify here the order in which you plan to implement the subcomponents of your system and the order in which you plan to integrate such subcomponents and test the integration.
7. **REFERENCES**

5.3 Deliverable 3 - Implementation

This deliverable includes all the sources of the software system coherent with the requirements analysis document and the design document. It should include the set of functions that you consider to be essential for engaging the identified users. We do not expect a market-ready product, but a viable prototype. Nice-to-have features could be disregarded in this implementation. Privilege stability and quality of your code rather than the implementation of a large quantity of features.

We do not impose any requirements on the programming language and frameworks/libraries adopted for the implementation.

More in detail, this deliverable must include:

- The code ready for installation and execution
- Installation instructions: anything that is useful to install and run your software, any prerequisite, dependencies, assumptions.